FUN-5.A.1 The definite integral can be used to define new functions.

FUN-5.A.2 If f is a continuous function on an interval containing a, then $\frac{d}{dx}\left(\int_a^x f(t)dt\right) = f(x)$, where x is in the interval.

FUN-5.A.3 Graphical, numerical, analytical, and verbal representations of a function f provide information about the function g defined as $g(x) = \int_a^x f(t)dt$

6-4 & 6-5 The Fundamental Theorem of Calculus and Accumulation Functions

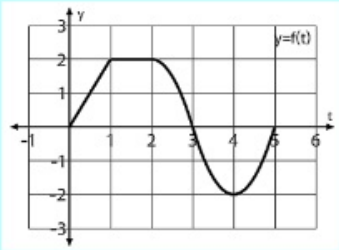
Get Ready...

We have learned about the definite integral and how $\int_a^b f(x)dx$ represents the area under the curve of $f(x)$ on the interval $[a, b]$.

Let's say we know the interval starts at a, but we don't know where it stops. That would give us the interval $[a, x]$ where a is a constant and x is some unknown variable. We can represent this new integral as a function that looks like this:

$$F(x) = \int_a^x f(t)\, dt$$

Let the function $f(t)$ be the graph at the right.



Let $F(x)$ be defined as the function $F(x) = \int_0^x f(t)dt$. Use the graph at the right to find the values of $F(0)$, $F(1)$, $F(2)$, $F(3)$, $F(4)$ and $F(5)$. The first two have been done for you.

$F(0) = \int_0^0 f(t)\, dt = 0$

(there is no area under the curve between 0 and 0)

$$F(1) = \int_0^1 f(t)\, dt = \frac{1}{2}(1)(2) = 1$$

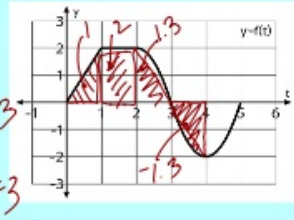
x	$F(x)$
0	0
1	1
2	3
3	4.3
4	3
5	1.7

$F(2) = \int_0^2 f(t)dt = 1+2 = 3$

$F(3) = \int_0^3 f(t)dt \approx 1+2+1.3 = 4.3$

$F(4) = \int_0^4 f(t)dt = 1+2+1.3-1.3 = 3$

$F(5) = \int_0^5 f(t)dt \approx 1+2+1.3-1.3-1.3$
$= 1.7$



The function $F(x)$ is represented by an integral and is called an **<u>accumulation function</u>**.

The Fundamental Theorem of Calculus (part 2) FTC	If f is a continuous function on $[a, b]$ and $g(x) = \int_a^x f(t)\,dt$ for all x in $[a, b]$, then $g'(x) = f(x)$ In other words: $\frac{d}{dx}\int_a^x f(t)\,dt = f(x)$

Derivatives and Integrals are **inverses** of each other. They cancel each other out, just like multiplication and division. In the "Get ready" problem, the graph of f is the derivative of $F(x)$. So $F(x)$ is considered the **antiderivative** of $f(\cancel{x})$.

AP Note: This is **guaranteed** to be on the AP exam. Think of this as another derivative rule to memorize!

Example 1: Find $F'(x)$, if $F(x) = \int_2^x \sqrt{1+t^2}\ dt$

$F'(x) = \sqrt{1+x^2}$

Example 2: Find $F'(x)$, if $F(x) = \int_{-1}^{x} \underbrace{\sin(2t+5)}_{f(t)}\, dt$

$F'(x) = \sin(2x+5)$

Three Important Notes about the Fundamental Theorem of Calculus

1. The FTC (part 2) is what we will use when we need to differentiate a definite integral with variable(s) in the limits.
2. The FTC (part 2) allows us to differentiate a definite integral WITHOUT integrating or differentiating the integrand itself.
3. Since we are ultimately taking a derivative, we will use chain rule and multiply by the derivative of the limit of the integral!

What happens when the upper limit of the integral isn't just x? **<u>Use the chain rule!</u>**

$$\frac{d}{dx}\int_a^{g(x)} f(t)\,dt = f\big(g(x)\big)\cdot g'(x)$$

Example 3: Find $g'(x)$ if $g(x) = \int_1^{x^4} \sec t \ dt$

$$g'(x) = \sec(x^4) \cdot 4x^3$$
$$= 4x^3 \sec(x^4)$$

Property for Integrals	$\int_a^b f(x)\,dx = -\int_b^a f(x)\,dx$

When finding definite integrals using graphs and geometric area, we always want the **smaller** number as the bottom limit of the integrand. This allows us to move forward and find area normally.

Example 4: Find $g'(r)$ if $g(r) = \int_{3r^2}^{5} \sqrt{1+5t}\, dt$. Be careful!

$$g(r) = -\int_{5}^{3r^2} \sqrt{1+5t}\, dt$$

$$g'(r) = -\sqrt{1+5(3r^2)} \cdot 6r$$

$$= -6r\sqrt{1+15r^2}$$

Example 5: $\frac{\mathrm{d}}{\mathrm{dx}}\int_{\sin(2x)}^{-2} e^{t^2}\ dt = \frac{d}{dx}\left[-\int_{-2}^{\sin(2x)} e^{t^2}\,dt\right]$

$= -e^{(\sin(2x))^2} \cdot \cos(2x) \cdot 2$

$= -2\cos(2x)\, e^{\sin^2(2x)}$

Example 6: (**2009 AB27**) If f is the function given by $f(x) = \int_4^{2x} \sqrt{t^2 - t}\, dt$, then $f'(2) =$

(A) 0 (B) $\frac{7}{2\sqrt{12}}$ (C) $\sqrt{2}$ (D) $\sqrt{12}$ (E) $2\sqrt{12}$

$f'(x) = \sqrt{(2x)^2 - 2x} \cdot 2$

$f'(2) = 2\sqrt{16-4} = 2\sqrt{12}$

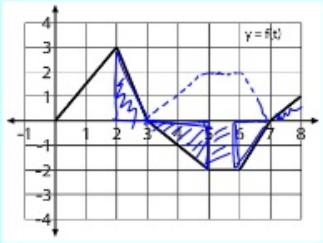
Example 7: Find $F'(x)$ if $F(x) = \int_{2x}^{3x} (t^2 - t)\, dt$

$$F'(x) = \left((3x)^2 - 3x\right)3 - \left((2x)^2 - 2x\right)2$$
$$= (9x^2 - 3x)3 - (4x^2 - 2x)2$$
$$= 27x^2 - 9x - 8x^2 + 4x$$
$$= 19x^2 - 5x$$

Day 2:

Get Ready...

Given the graph of $f(t)$ at the right, determine the value of each integral.



A.) $\int_2^5 f(t)\,dt = \frac{1}{2}(1)(3) - \frac{1}{2}(2)(2)$

$= \frac{3}{2} - 2 = -\frac{1}{2}$

B.) $\int_2^8 |f(t)|\,dt = \frac{3}{2} + 2 + 2 + 1 + \frac{1}{2}$

$= \boxed{7}$

Let's combine accumulation functions of the form $\boxed{g(x) = \int_a^x f(t)\,dt}$ with what we learned in Unit 5, where we analyzed the first and second derivatives to understand the behavior of a function. $g'(x)=f(x)$

Behavior of Accumulation Functions		
g(x) has/is...	**When g(x)...**	**Or when f(x)...**
Increasing	$g'(x) \geq 0$	$f(x) \geq 0$
Decreasing	$g'(x) \leq 0$	$f(x) \leq 0$
Relative minimum	$g'(x)$ changes from $-$ to $+$	$f(x)$ changes from $-$ to $+$
Relative maximum	$g'(x)$ changes from $+$ to $-$	$f(x)$ changes from $+$ to $-$
Concave up	$g''(x) > 0$	$f'(x) > 0$
Concave down	$g''(x) < 0$	$f'(x) < 0$
Point of Inflection	$g''(x)$ changes signs	$f'(x)$ changes signs

Example 8: Let $f(x) = \int_1^x (2t^3 - 3t^2 - 12t + 7)dt$. Find the interval(s) where $y = f(x)$ is concave up. $f'' > 0$

$$f'(x) = 2x^3 - 3x^2 - 12x + 7$$

$$f''(x) = 6x^2 - 6x - 12 = 0$$

$$6(x^2 - x - 2) = 0$$

$$6(x-2)(x+1) = 0$$

$x = 2 \quad x = -1$

f'': $+ \quad - \quad +$

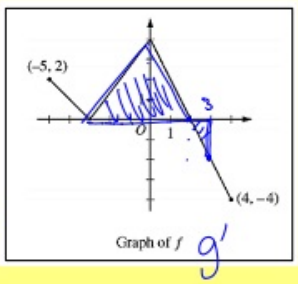
f: $\cup \quad -1 \quad \cap \quad 2 \quad \cup$

f is concave up: $(-\infty, -1)$ and $(2, +\infty)$

Example 9: 2014 AB3 (Graphical)

The function f is defined on the closed interval $[-5, 4]$. The graph of f consists of three line segments and is shown in the figure at the right. Let g be the function defined by

$$g(x) = \int_{-3}^{x} f(t)dt \qquad g'(x) = f(x)$$



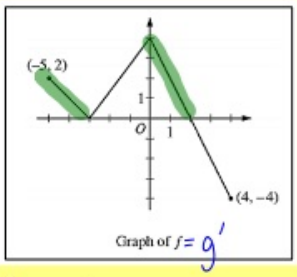
Graph of f g'

(a) Find $g(3)$.

$$g(3) = \int_{-3}^{3} f(t)dt = \frac{1}{2}(5)(4) - \frac{1}{2}(1)(2)$$
$$= 10 - 1 = \boxed{9}$$

(b) On what open intervals contained in $-5 < x < 4$ is the graph of g both increasing and concave down? Give a reason for your answer.

g inc $\to g' \geq 0$, $f \geq 0$ $\qquad$ g conc down $\to g'' < 0$ g' dec, $f' < 0 \to f$ dec



Graph of $f = g'$

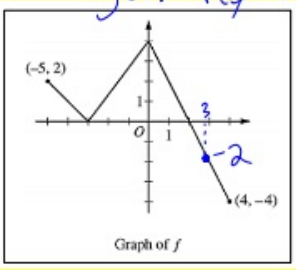
g is inc and concave down on $(-5,-3)$ and $(0,2)$, because $g'(x) = f(x) \geq 0$ and $g'(x) = f(x)$ is decreasing on these intervals

(c) The function h is defined by $h(x) = \dfrac{g(x)}{5x}$. Find $h'(3)$.

$g'(3) = f(3) = -2$

$$h'(x) = \frac{5x \cdot g'(x) - g(x) \cdot 5}{(5x)^2}$$

$$h'(3) = \frac{15 \cdot g'(3) - g(3) \cdot 5}{15^2}$$
$$= \frac{15(-2) - 9(5)}{15^2}$$
$$= \frac{-30 - 45}{225} = \frac{-75}{225} = -\frac{1}{3}$$



Graph of f

Day 3:

The Fundamental Theorem of Calculus (part 1)	If f is a continuous function on $[a, b]$ then $$\int_a^b f(x)\,dx = F(b) - F(a)$$ where F is any antiderivative of f, that is, $F' = f$

Big Ideas Review:

1. On the AP exam, we are often given a **definite integral** or **rate function** and are asked to find a **total amount** **or function value**.
2. This idea is called **accumulation** and is based on the **Fundamental Theorem of Calculus**.
3. These problems occur in many different **formats**:
   contextual problems, particle motion, graphs, equations, etc…

AP Exam Tip: Problem gives derivative (rate) and questions ask for definite integral (value).

The Net Change Theorem	The definite integral of a rate of change, F', is the net change in the original function F. That is $$\int_a^b F'(x)\,dx = F(b) - F(a)$$ Written another way: $$\underbrace{F(b)}_{\text{End Amount}} = \underbrace{F(a)}_{\text{Start (Given) Amount}} + \underbrace{\int_a^b F'(x)\,dx}_{\text{Net Change}}$$

Let's compare the two theorems:

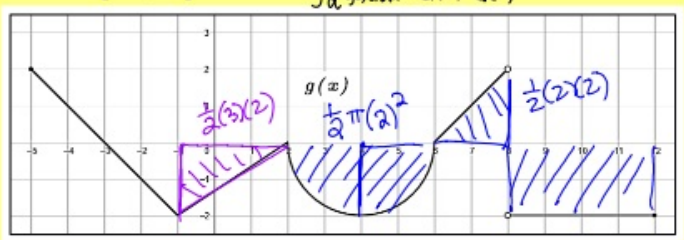
The Fundamental Theorem of Calculus (part 1)	**The Net Change Theorem (or Accumulation Model)**
$\int_a^b f(x)\,dx = F(b) - F(a)$	$F(b) = F(a) + \int_a^b F'(x)\,dx$

We've been doing these types of problems in context, but these types of problems can appear in many different ways!

Example 10:

The graph of $g(x)$ is shown above for $[-5, 12]$ and consists of four line segments and a semi circle. $G(x)$, the antiderivative of $g(x)$, is continuous on $[-5, 12]$.

$\int_a^b g(x)dx = G(b) - G(a)$



a.) Given $G(2) = 0$, find $G(12)$ and $G(-1)$

$G(12) = G(2) + \int_2^{12} g(x)dx$

$= 0 + -2\pi + 2 - 8$

$= -2\pi - 6$

$G(-1) = G(2) + \int_2^{-1} g(x)dx$

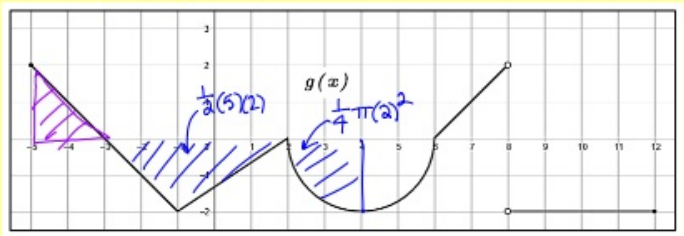
$= G(2) - \int_{-1}^{2} g(x)dx$

$= 0 - [-3]$

$= 3$

Example 10:

The graph of $g(x)$ is shown above for $[-5, 12]$ and consists of four line segments and a semi circle. $G(x)$, the antiderivative of $g(x)$, is continuous on $[-5, 12]$.



b.) Given $G(-3) = 5$, find $G(-5)$ and $G(4)$

$$G(-5) = G(-3) + \int_{-3}^{-5} g(x)\,dx$$
$$= G(-3) - \int_{-5}^{-3} g(x)\,dx$$
$$= 5 - [2]$$
$$= 3$$

$$G(4) = G(-3) + \int_{-3}^{4} g(x)\,dx$$
$$= 5 + -5 - \pi$$
$$= -\pi$$

Example 11: $f'(x) = \sin^2(3x - 7)$. If $f(5) = 9$, find...

a.) $f(8)$

$f(8) = f(5) + \int_5^8 f'(x)\,dx$ ← on calc.

$= 9 + 1.431917\ldots$

$= 10.432$

or

10.431

b.) $f(-3)$

$f(-3) = f(5) + \int_5^{-3} f'(x)\,dx$

$= 9 + -3.978039\ldots$

$= 5.022$

or

5.021

Example 12: $g(x) = \ln|x^2 + 3x + 2|$. If $G(-3) = 6$, find...

a.) $G(4)$

$G(4) = G(-3) + \int_{-3}^{4} g(x)\,dx$

$= 6 + 6.18404\ldots$

$= 12.184$

b.) $G(-7)$

$G(-7) = G(-3) + \int_{-3}^{-7} g(x)\,dx$

$= 6 + -9.411452\ldots$

$= -3.411$